Exploring Hybrid Virtual-Physical Homes

Lauren Herckis[1]
lrhercki@andrew.cmu.edu

Jessica Cao[1]
jkcao@andrew.cmu.edu

Jacqui Fashimpaur[1]
jfashimpaur@gmail.com

Anna Henson[1]
henson.anna@gmail.com

Rachel Rodgers[1]
rachel.rodgers42@gmail.com

Thomas W. Corbett III[1]
tcorbett@andrew.cmu.edu

Jessica Hammer[1]
[1]Carnegie Mellon University
hammerj@andrew.cmu.edu

ABSTRACT

There has been a great deal of research regarding the ways that people conceptualize and interact with places they call "home." This work has been extended to virtual reality primarily in the context of design tools. Here, we explore the idea of a hybrid virtual-physical home that uses virtual space to supplement an existing physical home. We identify variation in the spatial context of place attachment to homes. A user study (N = 25) revealed the expectations and needs these groups have for virtual home spaces, including security, familiarity, and community. We prototyped and playtested three virtual rooms, and identified design features that supported feelings of home. Finally, we present challenges to designing home-like VR environments that can shape future research in this area. As virtual reality grows increasingly accessible, these findings will inform the design of virtual spaces.

Author Keywords

Virtual reality; Augmented home; Hybrid home

CCS Concepts

•**Human-centered computing** → **Human computer interaction (HCI); Human computer interaction (HCI); Virtual reality;**

INTRODUCTION

Home is recognized as the "prototypical" place [22], a central symbol and organizing principle in peoples' lives. Homes, like other meaningful places, are created when a space becomes imbued with attachment, identity, value, memory, and practice [12, 24, 19]. Homes therefore do not only function *pragmatically*, as a place where people eat, sleep, store belongings, or relax, but also *affectively*, as spaces where people experience privacy, security, and safety, and *culturally*, as spaces where they construct a sense of self, family, and community.

Not all people have equal access to all of these facets of the home. To a person experiencing domestic abuse, for example, the home may not be a place of privacy or safety. People may not be able to use the home expressively if they are renting an apartment, or if they are living in a situation where they cannot be open about their identity (e.g. a trans person who is concerned about disclosure). People who move house on a regular basis, such as students or military families, may struggle with familiarity and with a sense of ongoing care for a space. Considering home as an explicitly physical space reveals additional limitations around the amount of space in the home (e.g. people who are downsizing), changing landscape around the home (e.g. people living near eroding coastlines or expanding urban sprawl), and the home being fixed in a single location (e.g. people who travel extensively).

External pressures may also unexpectedly alter people's relationship to their homes. For example, we wrote this paper prior to the Covid-19 pandemic, and edited it for publication while under quarantine. This experience has, for many, brought forward existing challenges around the home. For example, people in small apartments may struggle with a lack of outdoor access, working remotely without an home office makes clear the benefit of a professional setting and access to work resources, and those quarantined with small children (such as several of the authors) face ongoing challenges around privacy.

In this paper, we explore the concept of a hybrid virtual-physical home as a response to these challenges. The hybrid home uses virtual spaces to extend and supplement existing home spaces, and to address problems of the home that the user cannot necessarily address in a home's physical space. For example, imagine being able to capture and revisit your childhood bedroom, or having a private home space for large art projects as an addition to your physical home.

To better understand the potential for hybrid home experiences, we conducted interviews to explore peoples' relationship to home both as a a *physical space* and home as an experiential and *socially constructed place*. We identified three key themes: security, community, and familiarity. We then created design probes - sample virtual-physical hybrid home spaces - that would allow us to explore how these themes might be instantiated in the hybrid home context. We playtested these experiences with users and draw out design lessons for designing the hybrid home. We identify the home as a location from which to access virtual realities, and virtual reality as a way of extending home spaces and places. We also identify virtual reality as a way of accessing a physically distant home. We

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

DIS 2020, July 06–10, 2020, Eindhoven, NL

© 2020 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ISBN 978-1-4503-6974-9/20/07…$15.00

DOI: http://dx.doi.org/10.1145/3357236.3395561

conclude that extending an existing, physical home through the addition of virtual spaces entails novel design considerations.

This paper contributes the following: 1) a design vision for the virtual-physical hybrid home, 2) a discussion of variation in spatial place attachment to homes, 3) key themes for home-based hybrid experiences, and 4) design insights for creating security, community, and familiarity in the hybrid home.

RELATED WORK

Homes as spaces and places

Harrison and Dourish introduced the idea of a *place* as a special kind of space, one which has been invested with understanding, values, and expectations. They observed that, while spaces can be designed, places must be constructed by people over time through interaction. In other words, it takes time for a place to feel like *your* place, or for a new house to feel like home. The spot where you and your friends regularly gather becomes a meaningful place because mixing together the right people, in the right space, enough times is a recipe for an emotional whole that is more than the sum of its parts. These socially and culturally constructed places in turn frame appropriate behavior in relation to, and shape expectations of, that place [19]. In revisiting the distinction between space and place more recently, Dourish explores how emerging technologies such as GPS and wireless networks create opportunities for computer supported collaborative places which are not tied to localized physical spaces, and identifies some of the challenges inherent in placemaking in virtual spaces [12].

A home is a special kind of *place*: a spatial anchor and a symbol of comfort, identity, and refuge [22, 13]. While the concept of home is often defined geographically and spatially, in terms of buildings, institutions, neighborhood, region, country, and more [11], human relationships are also an important part of the construction of home [23]. Easthope describes three broad categories that encapsulate what people want and need from a place they call "home": territory, continuity, and identity [13]. A home territory may evoke feelings of privacy and security [11], especially in relation to boundaries between a home or domestic sphere and a external or public sphere [40]. A sense of permanence or continuity is central to ideas of home, encompassing familiarity and ritual associated with a place [11]. Finally, through personalization of, or a person's adaptation to, a space, people may build their identity into and around a home [11]. The home may become a symbol of the self [23] and a way to distinguish between self and other [13].

Place attachment, an emotional bond between people and their environment, has been widely studied but is still not well understood [22]. Anton and Lawrence [3] note that a place attachment to one's home has been linked to many positive health benefits and positive engagement with a broader community and environment, and that its lack is associated with stress and negative health outcomes. Some constructs related to place attachment, such as place identity, are not in fact anchored in space. These conceptual, emotional, or symbolic elements of place attachment make it possible to be in one place at one time, but interact with objects, items, or technologies which carry an experience of attachment to a place which is distant in space. Ylirisku et. al., for example, explored the design challenges in supporting active place presence at home [48]. A home can be understood as a foundational place in peoples' lives which is culturally, socially, and individually defined, and so the strength and nature of place attachment associated with the home varies with many factors [22].

The home in HCI research

Prior home-related work in HCI falls into seven main categories, according to Desjardins et. al. [10]: social routines, domestic practices, testing grounds for new technology, smart homes and automation, contested values of the home, sites for interpretation of technology used beyond functionality, and speculative visions [10]. Taylor et. al. address the interplay between smart technologies and the strategies people enact on, in, and in concert with elements of their home [44], and the Aware Home Research Initiative at Georgia Tech has been conducting research that addresses the future of domestic technologies for two decades [21].

Of these approaches, we are particularly interested in how technology supports home-based placemaking. Csikszentmihalyi and Rochberg-Halton found that domestic objects are often more valued for the meanings ascribed to them by people–their style, the memories associated with them, and so forth–than for their utility [34]. Odom, et. al. conducted a user study about teenage interaction with different digital artifacts presented in their physical home-space [29]. Sellen et. al. explore coordination and communication within a residence through investigation of remote, situated messaging within the home [37]. Crabtree, et. al. conducted an analysis of information coordination in the home via mail [8], while Swan and Taylor examined photo displays [42] and organizational artifacts such as calendars [43]. They argue that the home is composed of groups of heterogenous artifacts that capture and communicate information, which suggests broadly that, digital artifacts designed for the home should be flexible and able to be integrated into established home-systems [43]. Ringland demonstrates that when identity extends into virtual places, placemaking and community-building imbue these spaces with meaning and can confer a sense of "home" [33].

Finally, the games literature has explored gameplay behavior in the home. These games are often theorized as an appropriate intervention for users to voluntarily engage in a target behavior at home, such as continuing physical therapy or raising activity levels [49]. By making the activity playful, it becomes home-compatible. Games also *represent* home spaces, such as the farm in *Stardew Valley* or houses that the player can buy in *Skyrim*. While players experience attachment to these virtual homes, the home in the game belongs to the *character* rather than the *player* [45]. While they might describe it as "my" home [17], in these cases the home ultimately exists within the frame of the game, rather than in the larger social experience of the player [15]. These in-game "homes", belonging to characters, are experienced as distinct from physical homes belonging to the players.

Mixed reality and the home

The primary research thrust around homes and augmented/virtual reality (AR/VR) attends to the home as a physical space, but not as a place imbued with emotional attachment. The most common approach is to use AR/VR as design tools in and for the home. Such design considerations range from Oksman's suggestion to use augmented reality (AR) to help users design homes more intuitively and interactively [30], to Ceccacci's use of AR/VR to prototype e-kitchen implementations to assist the elderly and disabled [5]. Chia and He explored how families make home-related decisions, and how they interacted with AR-enabled design tools to visualize home improvements and projects [6, 20]. Results of these studies focus on improvements to the home design process. For instance, Oksman found that 80% of people would find an AR home-design process useful [30]. Meanwhile, Chia and He focused on the dynamic of personal relationships in design, such as how people seek out the approval of their partners [6]. In their speculative design research, Oogjes et. al. propose an augmented reality solution in which users can filter out different objects in order to create a flexible home environment which meets changing needs [31].

The home has also been explored as a location in which to conduct user studies, as a context for accessing virtual and mixed reality, and as a context for augmented reality. Mottleson and Hornbaek found that performance on tasks related to pointing, tracing, and body ownership illusion is not affected by the location in which VR tasks are conducted; that is, users perform similarly whether the experiment is conducted in a lab or at home [27]. Simeone and Velloso explored the potential for substitutional reality in the home, and the complexity of integrating physical with virtual surroundings [39]. Ringland describes the intertwined nature of physical and digital environments for players [33]. Clark and Dutta networked smart homes to enable remote social interactions and support an illusion of shared space in the home [7]. Sra et. al. developed a series of spatial mapping techniques to understand co-presence among remotely located, virtually co-present users of room-scale VR in which the size and layout of their physical spaces differed. These physical-to-virtual spatial mapping techniques were found to enhance perception of togetherness and embodied co-presence [41].

Industry VR experiences have also explored the notion of home. These studies have posited virtual spaces serving some of the functions associated with homes. PlayStation Home was a shared VR space intended to give players their own apartments in a virtual space, that players could decorate to their liking. Users could meet in their digital Home and play games together, or meet in shared "clubrooms" [1] (though it never left beta, and officially closed in 2015 [32]). Netflix VR gives users the option of watching Netflix in a virtual "cozy den" [46]. SteamVR Home allows for customization of users' space and avatar, as well as providing the ability to share these spaces with other users [2]. While little research has been conducted on these experiences, they evoke home through analogies to physical spaces, and are not related to or integrated with other places the user may experience as home.

In this research we explore the feasibility of hybrid physical-virtual homes created by *extending physical homes into virtual spaces*. We investigate this innovative design concept through a set of design probes around virtual spaces that might be situated as extensions of physical homes. We additionally address design implications for developing hybrid physical-virtual homes. This effort enables future research on, and development of, hybrid physical-virtual homes that meet a distinct, and to date unexplored, set of needs.

METHODS

This project builds upon prior work on the home and probes home-like aspects of virtual spaces. We combine three approaches to understanding the space: 1) semi-structured interviews, 2) design probes, and 3) playtest sessions with our probes, which included additional post-play interviews.

Participants

Twenty-five participants aged 18 to 63 were recruited through snowball sampling, flyers posted in various city locations and on two university campuses, and a Craigslist ad. Participants were compensated with Amazon gift cards at a rate of $15 per hour and, to ease recruitment, were required to be at least eighteen years of age and affiliated in some way with one of five postsecondary institutions. Unlike many studies that only recruit students, we worked with staff, faculty, and other university populations, who are diverse in race, socio-economic status, age, and other factors. Of the 25 participants recruited, 12 were 24 years of age or younger, 7 were between the ages of 25 and 39, and 6 were 40 years of age or older. Twelve participants identified themselves as female and 13 as male. Participants were offered the choice of interviews about the concept of home, playtests, or both; 6 participants chose to playtest only and were therefore only debriefed about their experiences with the design probes.

Semi-Structured Interviews

Members of the research team conducted a set of nineteen semi-structured interviews lasting between 35 and 75 minutes, performed in private meeting or lab rooms. We began by explaining the purpose of the study and our guarantee of confidentiality, and obtained consent from all participants to record. Participants were prompted to share specific ideas about "home", as well as to identify the characteristics, needs, and experiences central to their understanding of "home." We probed on themes from prior literature, including concepts of comfort, ritual, memory, and identity. Our interview process also allowed us to explore concepts raised as important by participants. These interviews were completed before the design process, and directly informed the creation of our probes.

Interviews prompted participants to express their ideas of home, and to explore physical sensations both in virtual experiences and in experiences of home. Prompts explored definitions and organization of home, as well as themes of continuity, territory, ritual, temporality, comfort, and privacy. Interview transcripts were reviewed and analyzed through a structural coding approach [14], which entails multiple rounds of open coding in a grounded theory framework to produce insights [9]. We refined these themes via axial coding [35], in which

emerging insights were revised and analytic connections were then made across coded categories.

As part of the interview process, facilitators provided plain paper (larger than A4), a box of 64 crayons, an assortment of colored Sharpie markers, and a black pen. Participants completed a brief written questionnaire, engaged in a face-to-face interview, and were then asked to create concept maps on paper to illustrate significant characteristics of "home" [4]. All concept maps were photographed and stored digitally and analyzed qualitatively through an examination of characteristics including inclusion, connection, centrality, proximity, and degrees of connectedness between concepts [36].

Design Probes

Design probes can inspire participants to respond to elements and relationships in surprising and informative ways [16], and have been widely and variably adopted into HCI research. Our probes were used as both tools for design and tools for exploration, as per Wallace et al [47]. We follow Mattelmaki [25], however, in using probes to inform designs around the broad concept of hybrid virtual-physical homes; we did not collaborate with individuals to design hybrid home experiences especially for them. Our probe designs can therefore be understood as hypotheses informed by the results of our semi-structured interviews, and also as design ideas which we developed to purposefully direct participant exploration and articulation of complex, personally held ideas about the home.

Central themes emerging from the semi-structured interviews included security, familiarity, and community. To explore these organizing concepts, we created three virtual rooms to use as design probes: the Meditation Room, the Greenhouse, and the Hearth. Each room was created in Unity and built for the HTC Vive. The Greenhouse incorporated Leap Motion technology, while the Hearth incorporated physical props. (For more details of the probe experiences, see the design section below.) Rooms were created using an iterative design process, with each room taking approximately three weeks to produce. Once a preliminary version of the room was complete, pilot subjects provided preliminary feedback to help make the room more usable, and to prepare the experience for playtesting. This process enabled us to discover what sense users made of their experiences in these rooms, and how they might extend their unique needs around home into virtual spaces. These designs emerged from the interview process, and were chosen because the design team felt they could communicate the core themes we were exploring through those lenses.

In our designs, we sought to have participants extend their attachment to real-world home places into the virtual, instead of being attached to the virtual space independently. In our early designs, we explored a range of approaches, such as creating fantastical or impossible spaces. However, we found that in order to project their own attachments into a space, participants needed a blank canvas, so to speak, onto which they could imagine their own experiences. The designs outlined below follow this strategy of building spaces that evoke familiar signifiers and support participant projection.

Playtests

All playtests were conducted in a private lab space on campus. At the beginning of each playtest, users were introduced to the study and given information about the room they would enter before donning the headset. Participants were informed about the interactions available in each space, but were not asked to complete specific tasks. For example, before entering the Greenhouse, participants were told, "This is a Greenhouse where you can grow your own plants. You won't be using the controllers here; just your hands. You can weed, you can dig, you can plant seeds..." Members of the research team recorded observations about participant behavior, including both stated and apparent emotional responses to the environment, activities within the virtual environment, and challenges encountered by participants.

Participants were recruited on a rolling basis, and design probes were playtested as they were developed. Playtests of the Meditation Room (n=17) and the Hearth (n=10) included one round of play; for the Greenhouse (n=19), participants experienced the space repeatedly, either within a single session or across multiple playtest sessions. After completing play, a member of the research team asked participants a series of exit interview questions addressing sensations, impressions, and hypotheticals. Questions about sensations addressed the emotional impact of the room as a whole (e.g. "How comfortable did you find this space?"). Questions about impressions asked people to respond to specific elements of the room (e.g. "How did the ability to change the time of day affect your experience?"). Hypotheticals prompted participants to imagine potential use cases (e.g. "What can you see yourself doing in this room?", "Imagine this room could accommodate multiple users at once. Who would you want to share it with?").

Audio and video from playtests were integrated with observational notes and exit interview data, and were then qualitatively analyzed in three passes. First, playtest data for each room was reviewed for evidence of technical and design challenges as well as preferred modes and loci of engagement. Next, playtest data was analyzed as a corpus using an open coding approach for content related to the themes identified through literature review and semi-structured interviews, including continuity, territory, ritual, comfort, privacy, and identity. Finally, an axial coding approach was used to explore thematic and experiential differences in playtest data across rooms and user categories.

INTERVIEW FINDINGS

Participants described their homes in many and varied ways. Some described physical structures, soundscapes, and people in great detail. For many, home is the place where they keep meaningful objects and enact meaningful relationships, and most participants who said that they feel far from home said that they did not want to stay away for long. Many described specific memories or ritual behaviors, such as a place where they connected with loved ones, learned to cook signature dishes, keep their treasured mementos, or take refuge from a real or metaphorical storm. Still others described their homes in entirely spaceless terms: they focused on rituals, objects, and people, which could be encountered in different spaces.

Familiarity

Participants cited familiarity as a source of comfort. P05 said that, because he had lived in one place "for nine years, I was very comfortable with everything that was there. I kind of knew it inside and out and the fact that I kinda got to call it my own." When describing physical homes, participants referenced adding objects or the "personalization" (P37) of space through alteration, such as by painting. You can make a new space a home, suggests P38, by "adding your own furniture, you know, having something that is familiar that you can add to an unfamiliar space." Learning a space and its contents, or repeated interaction with the space, affords a sense of familiarity. Says P01, "We have a new couch now that doesn't feel like home because it's just new foreign object."

This theme of *familiarity* also manifested among participants who related to home spacelessly. They focused on objects or relationships, but still described home as a locus for autonomy, comfort, and control. P06 exemplifies this experience when he says, "[I had an] experience of backpacking for six months and I had a very, very limited number of things, and so just feeling them, looking at them, setting them up everyday, taking them apart, like that routine. That was very much a home kind of feeling, just, you know, I'm unloading my backpack and just looking at the things in it. And even though there was such a small number of them, that was more of a feeling of being at home even though it was moving constantly in a different place every day. That was more of a feeling of home than, let's say, like the apartment I live in now."

Community

While we asked participants to describe their *personal* experience of home, a large number of participants talked about home as a *shared* physical place. For example, P36, above, explains that "we" chose the items in the home. Her experience of home is not just about the sense of familiarity, but also about the people with whom she shares it.

Many participants described social interaction, important relationships, or co-presence as central to their experience of home. P13 said, "I'm married and my wife, wherever she is would be home," and P22 said that home is "less about the zip code than who I'm around and feeling at ease." In other words, home could be a place regardless of space. When participants described an estrangement from a meaningful home place, they often described it in terms of estrangement from *community*. These participants included college students far from their family, individuals separating from romantic partners who felt a new estrangement from a place they once felt comfortable, and young professionals who had recently moved from a childhood home to a new space of their own.

Security

When participants were asked to describe their homes, most responded immediately with some variation of "a place of comfort and safety" (P13) or "a place where you feel comfortable and safe" (P17). Detailed descriptions of the security of home were notably sparse. To learn more, we turned to portrayals in concept maps, where "safe" and its variants were closely associated with "comfort", "safety", "privacy", and "relaxation." This sense of freedom from imposition or interruption extended to an implied freedom from danger, illustrated by P35 who says, "Having a dog makes me feel safe."

The security of "home" is not only a matter of freedom from external dangers, but also a sense of being able to let one's guard down, and of release from social niceties where one is free to "go, close the door, and be myself" (P22). Home can be "a sanctum sanctorum... my space, my retreat" (P33). P12 described home as a place where you "do what you want" and "don't do the things you don't want," wherever that may be.

DESIGN

As the quotes above suggest, we identified user needs around home both through elements that participants *currently had* in their relationship with home, and in elements that they *lacked* [3]. For example, participants who shared a home with others laughingly acknowledged the trade-offs of their entangled lives. They emphasized both the comfort and reward that can accompany this closeness, as well as the lack of individual autonomy or privacy.

Additionally, experiences of familiarity, community, and security are intertwined. For example, P10, a college student, describes her parents' home, saying, "I've lived there my whole life. So that's *home* home. Because this [residence at school] is constantly changing." The familiarity of living there her "whole life" is contrasted to the temporary nature of her residence at school. Similarly, P01 said that "once we cross the threshold we're kind of in our own safe little space." Home is a space to share with *desired* others, allowing denizens to experience a sense of security.

In our design probes, we responded both to the key themes from our interviews, and to the way that themes connected for our participants.

Design Probe One: The Meditation Room

The Meditation Room explores concepts of familiarity and security. It puts the user on a wooden deck overlooking an infinite ocean, with linen-like curtains on each side. There is a blue yoga mat in the center of the floor, and in front of that is a wooden box that serves as a diegetic user interface. By moving small rocks into two compartments, the user can adjust the volume of music and of sound effects such as splashing water. By sliding a particular rock back and forth, the user can adjust the virtual time of day from morning to sunset to night. There is also a small candle visible in the wooden box. All interactions are accomplished with the Vive controllers.

First, we explored familiarity through continuity in the Meditation room. Specifically, we examined whether elements of the physical world can "carry over" to the virtual space. To create a ritual of entry, we placed a yoga mat on the floor. Participants stood in the middle of the mat before putting on the headset. To begin the experience, users sat on the yoga mat; only once they were seated would the Meditation Room load. The mat was visible to users both in the physical and the virtual space. Second, we explored familiarity through repetition. We chose a room associated with a specific activity,

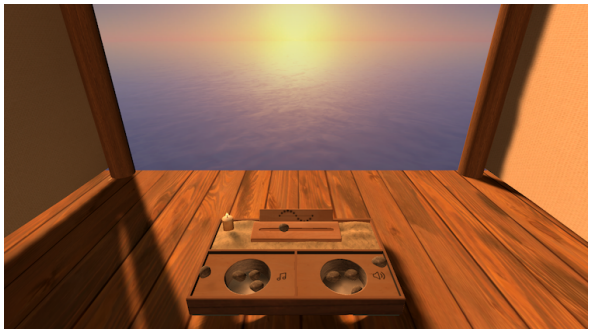


Figure 1. The Meditation Room, with the time of day set to almost sunset. ©Jessica Hammer



Figure 2. A view of the Greenhouse from the table. ©Jessica Hammer

meditation, in order to tap familiar activities in generating a sense of familiarity. To build on activity-based familiarity, the appearance of the meditation room was informed by design elements drawn from many real meditation rooms. Finally, we used spatial design to explore questions of security. For example, curtains were used to make the room feel small and private, without feeling cramped or confining.

Design Probe Two: The Greenhouse

The Greenhouse explores concepts of community and familiarity. Three walls are made of panes of glass in a green metal frame, through which a simple backyard with a large tree is visible. Inside the room, along each glass wall, are two rows of planters. The nearer row is subdivided into several "plots" of dirt, each about two feet by two feet. A wooden table along the fourth wall holds a watering can, a basket, and a tray of seeds. Above the table hangs a row of mailboxes, each with a name on it. There is ambient audio of birds chirping, leaves rustling, and other nature sounds. All interactions in this room use Leap Motion, which tracks hand position. Users can pull weeds from planting boxes, shoo away insects, dig holes in the dirt, plant seeds, pick fruit, put objects in the mailboxes, and water seeds or plants with the watering can. The user's hands are visualized as a pair of gardening gloves.

Mailboxes were the principle avenue for probing community. Hanging above the supply table in the Greenhouse are several wooden boxes, each labeled with the name of a fictional "neighbor." Every time the user visits the room, some of these boxes have letters or gifts, in the form of seed starters, from these neighbors. If a user leaves a plant in one of these boxes at the end of the day, it will be "sent" to the appropriate neighbor and future correspondence from them may reference the user's gift.

For familiarity, we were interested in exploring building place attachment through repeated exposure and control of the space. Playtesters experienced repeated exposure to the room through multiple entries and exits. Repeated playtests of a design probe over a relatively constrained period of time cannot create the same opportunity for attachment that multiple engagements with a place over many hours or years might. However, this approach enabled us to explore anticipation of, and retrospective ideas about, a space that playtesters returned to multiple times. Visits were separated by other playtests or by an interview. After each visit, we asked what they intended to do first the next time they visited the room. Between visits, any plants that were watered, weeded, and without bugs advanced one stage in their growth. Plant growth not only shows the passage of time in the virtual space, but also gives the user a sense of control. Playtesters could choose which seeds to plant, how to care for them, and what to do with any fruit that grows. Every user can customize their garden.

Design Probe Three: The Hearth

The Hearth explores concepts of security and community. Playtesters appear to be in an archetypical living room with bare walls, bookshelves, a wooden floor, a large rug, and a stone fireplace on one wall with a fire burning in it. On the wall to the right of the fireplace is a large window, which looks out onto pine trees and falling snow. Opposite the fireplace is a bookshelf, behind which a small dining area is visible. There are four furniture items in the middle of the room: two identical blue armchairs, a yellow ottoman, and a small table.

The Hearth was designed to explore aspects of security in a virtual space by evoking feelings of coziness and trust. Most design decisions in the room, including the fireplace, warm color palette, and winter scene visible through a window, were based on locally recognizable ideals of coziness. During playtests, we aligned a pair of large, cushioned armchairs and a padded ottoman with their virtual counterparts so that users could assume a comfortable position. We placed a space heater in the room, roughly aligned with the fireplace, and lighted a woodsmoke scented candle to stimulate users' other senses. Playtesters could sit in the armchair, put their feet on the ottoman, and rest an arm on the table. No interactions using either the Vive controllers or the Leap Motion were available.

We wanted to understand the role of physical feedback in user security. The furniture in the Hearth was modeled after real furniture placed in the room in precise alignment with their virtual counterparts. Researchers helped playtesters put on a headset, while other members of our team arranged furniture appropriately. Playtesters could sit in a virtual chair, and a physical chair would be there to catch them.

We also explored community through the concept of a mutually lived-in virtual place. The Hearth was designed to appear as though it were a small portion of a larger residence. In a dining area behind the armchairs, a door to another room admits light shining from an apparent space beyond. By the fireplace, an open door leads to a hallway with a visible bench. The bench



Figure 3. The furniture in the Hearth, as arranged in physical space (left) and as seen from within the headset (right). ©Jessica Hammer

has boots under it and a sweater hanging over it. These spaces suggest that the user is not alone, but rather has companions in the house with them. The second armchair contributes to this impression, implying that a second person might arrive.

PLAYTEST FINDINGS

Familiarity

One way users experienced familiarity with our design probes was through *continuity with known experiences*. We found that, in virtual spaces, continuity with the known affords a sense of familiarity and associated comfort. For example, users preferred the physical yoga mat and the virtual yoga mat of the Meditation Room to occupy the same space; when the mats were misaligned, participants went out of their way to align them. The mat made participants feel "more connected to what was happening" because they could "look down and see the mat [in the virtual space] and ... feel that I was on a mat [in the physical space]" (P22).

Players drew on real-world expectations to create a sense of continuity. In the Hearth, one playtester described liking it because "this place is probably in [the city where playtesting took place]" (P10), and another said, "It reminds me of similar evenings I've had where it's snowing outside" (P22). Familiarity was also produced through reference to recognizable tasks. In the Meditation Room, nearly half of the participants reported meditating in the space, while the rest were quiet and reflective. Participants who did not attempt meditation reported recognizing the room's purpose.

Finally, we explored the role of repeat exposure and routines in the development of familiarity. In the Greenhouse, some playtesters invented routines around inspecting new growth upon arrival or keeping the Greenhouse orderly before departure. Many people picked everything up off of the floor before leaving the space. Some participants expressed frustration when, due to implementation errors, certain items had to be left on the ground. Most participants made sure that all of their plants were in full health before leaving the room, and several reported feeling like these virtual plants were "theirs" (P17).

Community

Our design probes explored community by *suggesting* the presence of others rather than by *introducing* others into the space. We observed that participants were far more engaged by designs that suggested shared space rather than designs that focused on asynchronous communication. In the Greenhouse, many playtesters did not read letters from their fictional neighbors, and many who did "don't really remember what they said" (P10). On the other hand, participants reacted to suggestions of shared space in the Hearth, such as a second chair inviting company; architectural elements alluding to larger, unplayable areas; and apparent mementos on shelves. P22 explained, "It really makes me want to explore, you know, like when you go to someone's house for the first time and you want to peek around to see all their stuff!"

When it came to *who* they wanted to share space with, participants described people with whom they had prior, meaningful relationships. For example, participants wanted to "take care of" the Greenhouse with friends (P27), or to "have a nice talk in front of the fire" with family members in the Hearth (P22). Multiple participants said they wanted to interact with real people, not avatars or characters. For example, P06 said that he wanted to interact with "another person," not "some purple dinosaur kind of fake character." Researchers were included in these positive relationships. When we asked users if we might sit and chat with them in the Hearth, no one denied our request. Though they could not see us, many playtesters turned to face our chair during the conversation, which they had not done when speaking to us from within other rooms.

Security

We observed that playtesters evaluated their security through awareness of their virtual surroundings. For example, P22 described how the snowstorm outside made her feel "more comfortable" relaxing in the Hearth. P36 agreed; the "small cozy space" was "nice" because she could look out and feel that she was not "trapped in a very small space with cold weather outside." In the Meditation Room, playtesters described feeling safe "once you understand what's behind you" (P27). They also described being able to control the time of day, which affected their view out over the ocean, as contributing to "peaceful" feelings (P22, P34).

When participants were uncertain about their surroundings, they felt less secure. For example, some participants identified a half-open door in the Hearth as a source of anxiety. They did not feel secure because they worried that someone else might walk in, even knowing that the Hearth was a single-user experience. P30 said that if it were her space, she would have closed the door. Linking security to awareness of other people was often mediated through the concept of privacy: "I don't have a private space, but [The Meditation Room] made me feel like I was in one" (P22). However, participants also placed a high value on community and sharing. We identified *predictability* and *control* as elements that mediate participants' experience of sharing the space. Participants felt secure when they had clear expectations for social and behavioral norms, and when they had trusted relationships with co-participants. P22 described the Hearth as "a safe, easy environment to have a nice conversation" with her mother.

Activities in the VR-Home

Users overwhelmingly recognized the potential of virtual for their own lives, and often spontaneously offered use cases, even before being asked. We report the three most common themes here, but we believe that there is more work to be done in identifying VR-Home-relevant activities.

First, participants identified the VR-Home as a place to engage with intangible mementos such as digital photographs. Existing repositories were vast, scattered, and disorganized. They also could not easily be displayed in the users' physical homes. P13 agreed that her photographs were "in way too many different places right now, and so I'm doing less than I used to with them." P28 said, "It's hard to go back and look at them because there's just not enough time.... but there's just too much [data]." Individuals without a home place, in particular, described the pragmatic need to pare down their physical belongings. They would be glad to have virtual access to a consistent space where digital artifacts can "reside." For example, P37 suggested that the Hearth could be used "as a way to sit and chat about a digital photo album."

Second, playtesters repeatedly raised existing, real-world relationships. A number of playtesters suggested that the Hearth would be a great context for long-distance communication with loved ones who they don't often see in person. For example, P27 said, "I would like to share it with long distance friends, if I could use it as a sort of meeting space with friends who I can't see personally." P02 said, "The people make the place, that's my firm belief... you can have the biggest house, you can have even the smallest apartment, but if you don't have the people then it's not really a home." Participants whose descriptions of home were least spatial had particularly strong reactions to the VR-Home as a place where they might connect and reconnect with loved ones who are physically far away.

Finally, participants expressed a desire to use the VR-Home as a locus for emotional attachment to a place. We observed this desire primarily due to our inability to meet this need, as multiple playtesters described a lack of attachment to the virtual rooms they spent time in. This lack of attachment is appropriate given the short period of engagement with each room. Some remarked explicitly on their lack of attachment to the rooms, with P10 saying, "It just doesn't really feel like *my* space. It doesn't feel like some place I live in, but it feels more like just, if I'm over at somebody else's house or something." This lack offers new opportunities for design.

DISCUSSION

Recognizing the paramount importance of "home" in personal and social landscapes of meaning, we set out to explore the concept of a hybrid virtual-physical home, including a deeper understanding of the ways that experiences in virtual-physical hybrid homes might address user needs. Based on this work, we identified design insights for creating security, community, and familiarity in VR-home.

Virtual and Hybrid Home Spaces

In playtests and interviews, it became clear that people experience spaceless place attachment to the idea of "home", and are intrigued by the opportunity for placemaking in virtual home spaces. The hybrid virtual-physical home rests on an understanding that place-making is a persistent human behavior. Physical homes can rarely meet all needs. As new technologies create opportunities for people to experience new kinds of computational spaces and spend time there, place-making will naturally follow [12]. Virtual reality makes immersive, interactive experiences in virtual spaces possible, which creates the possibility for hybrid home spaces that imitate, extend, supplement, and have continuity with real-world physical homes. Hybrid spaces can become meaningful places through engaged, persistent interaction [22], and so are objects of potential place attachment. This study demonstrates that the opportunity for placemaking in virtual spaces posited by Harrison and Dourish [12] extends to central places such as "home".

Why the focus on hybridity? Our fundamental approach is to augment, rather than replace, the homes that people are attached to. While we chose to explore home-like virtual spaces in this study, our findings suggest that the virtual components of hybrid homes do not have to imitate real spaces. A meditation room could be a valuable addition to an urban apartment with limited window access, for example, providing a new opportunity in an existing home to deepen attachment [12]. Experiences of familiarity, security, and community make a space feel like home, but these feelings can be evoked in a range of ways and are not limited to representations of home like those belonging to in-game characters [45]. Virtual elements of a hybrid home could imitate physical home spaces as a means of evoking homeyness, and in our study we found that such imitation triggered familiar memories in players. However, we do not believe that such imitation is *necessary* to evoke a sense of home, or to extend an experience of home into virtual spaces. We recognize that our choice to focus on imitative spaces may limit both users' sense of what a space may be for, and also the ways in which our virtual spaces evoked affective and relational experiences.

Due to these design decisions, we encouraged playtesters to think of VR-Home as added rooms appended to existing houses. However, we might as easily imagine VR-Home creating portals *to* your home through virtual reality, *from* elsewhere. While our protocols did not account for this possibility, interview subjects described being able to "go home" from anywhere. Future research could explore a wider range of metaphors for connecting physical and virtual spaces, while still helping users feel safe [13], enact identity [11], and build relationships [23] within the boundaries of the home [40].

Finally, we note that even users experience embodiment simultaneously in physical and virtual environments [33]. Even when their attention may be in a virtual space, physical bodies exist in a physical location. There may be other people present in that location, and these people must be considered as part of the user's VR-Home experience. These people may be co-present in, or absent from, the virtual environment. In order to feel at home in a virtual environment, users must feel sufficiently secure in their physical environment. In the future, we expect to extend VR-Home to explore real-time interaction and virtual co-presence. This interaction and co-presence might include individuals who are co-present both physically and virtually, or who are present in one, but not both, of the physical and virtual environments.

Doing and Being

Participants' imagined uses for the VR-Home were *reflective* (engaging with personal media), *relational* (connecting with

others), and *emotional* (affecting their own emotional state). Participants did not envision using the VR-Home to further interpenetrate their home and work lives, e.g. as a quiet place to study or a place to conduct meetings. They also did not envision it as a space for domestic labor or caregiving, although the research team has identified possible opportunities for both. Instead, they saw the VR-Home as a place where they could reflect, rest, build relationships, play, and know themselves better. Activities of this kind are central to the placemaking that makes spaces into places [19].

We found that security and familiarity were mutually enhancing in playtest experiences. Surroundings that were familiar left playtesters feeling more secure. Familiarity was achieved by evoking prior experiences from other times and places, or by creating sensations which bridged the experiential transition between virtual and physical, such as when the appearance and texture of furniture was maintained in both the apparent virtual, and in current physical, surroundings of playtesters. Security and familiarity were also linked to control. The ability to manipulate aspects of the apparent environment helped playtesters to feel more secure within a space.

The Meditation Room was often cited as the most comfortable room because, as P22 said, "there were less stimulating things in the space ... I felt comfortable and just relaxed." While participants claimed that they wanted "something in the virtual world that I can do, like I could read a book or like draw or something in the virtual world" (P10), their behavior suggested otherwise. The Greenhouse was a particularly revealing locus for this experience, where some playtesters described the possibility of gamifying the experience. At the same time, however, user perception of the Greenhouse as a goal-oriented space introduced excitement, uncertainty, and unease. Playtesters tried to understand what was expected of them and grapple with the available options for completing tasks. Several participants actively explored spaces upon entry with an apparent goal of identifying manipulable elements of the environment and identifying the "right" thing to do. It seems likely that different results would arise from bespoke design probes which could explore an individual user's place dependence, a construct related to place attachment that comes from the experience of a place meeting an individual's needs or allowing them to achieve their personal goals [38].

Presence and Identity

Some playtesters valued the communal, social, familial, and collective aspects of their own sense of home, while others described individual experiences, needs, and values. This is consistent with our understanding of the home as a complex locus of place identity and place dependence. While homes can be places of privacy and solitude, they can also be places to interact with care and tenderness, where people are bound to one another [37]. Our design probes evoked communal living (the Hearth), solitary refuge (the Meditation Room), and the potential for both functions (the Greenhouse). Playtesters were intrigued by the potential of virtual rooms to serve in all of these capacities. We understand these findings through the lens of frontstage and backstage spaces [18]. People actively work to manage the boundaries between private and public in the home [28, 26]. This management process is accomplished through social interactions, through the arrangement of objects, and through the design of the physical space itself. For example, the rise of open kitchens in "transitional" homes brings the backstage work of meal preparation into the center of family life [28]. The physical limitations of the home, however, limit residents' ability to manage these boundaries. For example, guests may use a bathroom where private belongings such as medications and toiletries are stored.

In a VR-Home, people can have far more control over spaces that reflect different relationships or different levels of intimacy. For example, users might have one space for new acquaintances, another for family, and a third for close friends. One might think of this as a modern take on the distinctions between drawing room, parlor, sitting room, study, den, and more, that does not require residents to maintain many highly differentiated physical spaces. Users can also better control access to private spaces; for example, a guest cannot simply wander into one's bedroom while unattended in the VR-Home.

We alluded to other, unspecified people sharing a space, or suggested that no one else was nearby, as ways of exploring the key theme of community in hybrid virtual-physical homes. The presence of boots in an apparent hallway, or the assurance of a privacy curtain that no others were within view, provided a sense of community or a sense of solitude. In the Hearth, participants responded positively to the experience of "ambient presence," or the sense that other people shared the space even if they were not currently present. In particular, the experience of ambient presence was mediated by objects and belongings. In the Greenhouse, where interactions came from letters rather than a sense of ambient presence, participants were much less attentive to the relational aspects of the space. These insights are linked to a larger finding that awareness of apparent surroundings in virtual space was important. A number of playtesters remarked upon the door left ajar and the weather outside the window in the Hearth, and on the curtained enclosure and setting sun in the Meditation Room. Though we introduced characters to the Greenhouse, playtesters did not engage with these fictional people. However, playtesters did express a desire to share their virtual home experiences with community members, such as sitting by the Hearth with mom, or telling friends about thriving plants in the Greenhouse.

In addition to having fine-grained control over the design of the virtual space, users can also control how they present themselves in the context of different relationships. For example, users might express their gender differently in the context of close personal friendships compared to in public; they might have more freedom to express their gender in a virtual space compared to in their physical home, depending on who they live with. These decisions about identity expression can manifest in how the space is decorated, the affordances available to guests, the avatar used by the host, and more.

This work demonstrates that extending physical homes into virtual spaces is feasible. Playtesters found that our design probes offered the opportunity to attach, feel secure, and build community. Physical homes are limited by definition [13], and residents are limited by these contextual constraints. We build

on research into technology-supported, home-based placemaking to illustrate several ways that hybrid homes might extend both the footprint and the functionality of the home.

Homes where individuals lack access to physical space might be augmented with virtual rooms, like a balcony for a quarantined family or a lofty studio space for a sculptor, that offer this space. Families separated by physical distance might share a virtual space that's just through a doorway. Frequent travelers might have a familiar room that travels with them, and individuals who cannot express their identity openly might have a place they can be themselves. In other words, hybrid homes support pragmatic, affective, and cultural functions.

LIMITATIONS AND FUTURE WORK

The project was executed in a mid-sized city in the United States, which shapes the understanding of *home* addressed by this research. Future work will be needed to incorporate additional cultural perspectives. The equipment used for playtesting included a number of wire leads attached both to wearables and to stationary machines. The headset is adjustable but sometimes ill-fitting. Some virtual objects were challenging to grasp, manipulate, and release. These limitations likely affected the user experience, but will become less salient as VR hardware improves.

Expectations of, and experience with, virtual reality varies across populations. Some users were more comfortable with the physical equipment or navigating a virtual space than others. Though we avoided game elements like points, currency, and statistics, the Greenhouse conceptually resembles agricultural games like Farmville and Stardew Valley. Many participants used words like "play" and "game" when describing their time in the virtual rooms, a mindset that may have affected their experience. No playtester had previously experienced the addition of a virtual room to their physical home, and so none had prior experience with virtual reality as a place to extend their own emotionally layered experience of *home*.

Future research would profitably explore the role of different physical rigs, playtesters' prior experience with virtual reality, and the construction of virtual spaces which are tied to playtesters personal experiences of home. Playtesting hybrid experiences in users' own existing homes would shed light on best practices for integrating virtual spaces with physical homes and navigating within and between these spaces.

The current research explored single-user experiences. Many participants described a desire to extend meaningful personal relationships into specific virtual spaces, or wished for a virtual space in which they could extend meaningful personal relationships. Our project has the potential to meet user needs through the support of meaningful relationships regardless of distance and other contextual factors. Virtual rooms have the potential to become home *places* through the incorporation of multi-user experiences which allow meaningful interpersonal relationships to develop in virtual *spaces*. In the future, we will explore true multi-user experiences in the hybrid home.

CONCLUSION

In this paper we have presented the concept of a hybrid virtual-physical home, building on the home as both *place* and *space*. Through interviews and playtest sessions, we identified significant variation in place attachment to home and implications for extending the experience of home into hybrid spaces.

We identified key themes of security, community, and familiarity as being important for home experiences. We created design probes to explore how these themes might be instantiated in a hybrid virtual-physical space: a Meditation Room, a Greenhouse, and a Hearth. We learned about different design strategies for inspiring security, community, and familiarity. For example, participants responded much more strongly to environmental storytelling about community (e.g. seeing a pair of boots that aren't yours) than to asynchronous interactions with imagined community members (e.g. leaving letters for other gardeners).

By decoupling physical location from meaningful place, VR-Home makes it possible to build new home spaces which are not constrained by physical proximity or spatial constraints. Rooms for hybrid virtual-physical homes can be continuous with physical spaces located in different parts of the world, or completely independent of physical home spaces. With technological advancements that enhance the experience of co-presence in virtual spaces, hybrid virtual-physical homes could extend the ability of homeowners to share their home spaces with others. Place attachment anchored in a particular community and environment enhances positive engagement with that community and environment [3]. By decoupling home spaces from the constraints of cultural and social context, VR home could make it easier for individuals to withdraw from engagement with local community and environment. First, it may entice people to withdraw from real world social activities. Second, many relationships and dynamic communities are built because people must interact with diverse strangers in the course of their daily lives. By providing alternatives to real-world spaces, VR-Home could allow individuals to avoid some of the incidental social interactions which enhance and extend community engagement.

The experiences that we created do not directly target the problems that we identify as addressable with a hybrid virtual-physical home. For example, we did not explicitly create a space where someone can safely express their gender, or where they can reflect on a home they can no longer revisit in the real world. However, these design principles can help us create spaces that address people's emotional needs around home, and we can use them to create more tailored spaces. More importantly, we can use these principles to allow *users* to create spaces that meaningfully enhance their own experience of home, and to empower them to construct their identity, relationships, and roles using virtual technologies.

ACKNOWLEDGEMENTS

We thank Christian Guirnalda, Joshua Ness, and Zachary Drossman. This research was supported through grants from Oath, Inc.

REFERENCES

[1] 2014. Home Development Q&A. (2014). https://web.archive.org/web/20070320150512/http://www.scedev.net/home/Third%20Party%20Relations%20Q%26A.pdf

[2] 2017. Introducing SteamVR Home Beta. (2017). https://steamcommunity.com/games/250820/announcements/detail/1256913672017157095

[3] Charis E Anton and Carmen Lawrence. 2014. Home is where the heart is: The effect of place of residence on place attachment and community participation. *Journal of Environmental Psychology* 40 (2014), 451–461.

[4] Jessica G Burke, Patricia O'Campo, Geri L Peak, Andrea C Gielen, Karen A McDonnell, and William MK Trochim. 2005. An introduction to concept mapping as a participatory public health research method. *Qualitative health research* 15, 10 (2005), 1392–1410.

[5] S. Ceccacci, M. Germani, and M. Mengoni. 2012. User Centred Approach for Home Environment Designing. In *Proceedings of the 5th International Conference on PErvasive Technologies Related to Assistive Environments (PETRA '12)*. ACM, New York, NY, USA, Article 31, 8 pages. DOI: http://dx.doi.org/10.1145/2413097.2413136

[6] Foong-Yi Chia, Bokyung Lee, and Daniel Saakes. 2016. Collaboration in the Living Room or How Couples Design Together. In *Proceedings of the 9th Nordic Conference on Human-Computer Interaction (NordiCHI '16)*. ACM, New York, NY, USA, Article 99, 6 pages. DOI:http://dx.doi.org/10.1145/2971485.2996734

[7] Meghan Clark and Prabal Dutta. 2015. The haunted house: Networking smart homes to enable casual long-distance social interactions. In *Proceedings of the 2015 International Workshop on Internet of Things towards Applications*. ACM, 23–28.

[8] A. Crabtree, T. Hemmings, and T. Rodden. 2002. Coordinate displays in the home. *Proceedings of the 2002 ACM Conference on Computer Supported Cooperative Work (Workshop on Public, Community and Situated Displays), New Orleans: ACM Press* (2002). www.mrl.nott.ac.uk/axc/documents/CSCW_Displays.pdf

[9] Benjamin F Crabtree and William L Miller. 1999. *Doing qualitative research*. sage publications.

[10] Audrey Desjardins, Ron Wakkary, and William Odom. 2015. Investigating Genres and Perspectives in HCI Research on the Home. In *Proceedings of the 33rd Annual ACM Conference on human factors in computing systems (CHI '15)*, Vol. 2015-. ACM, 3073,3082.

[11] Carole Després. 1991. THE MEANING OF HOME: LITERATURE REVIEW AND DIRECTIONS FOR FUTURE RESEARCH AND THEORETICAL DEVELOPMENT. *Journal of Architectural and Planning Research* 8, 2 (1991), 96–115. http://www.jstor.org/stable/43029026

[12] Paul Dourish. 2006. Re-space-ing place: place and space ten years on. In *Proceedings of the 2006 20th anniversary conference on Computer supported cooperative work*. ACM, 299–308.

[13] Hazel Easthope. 2004. A place called home. *Housing, Theory & Society* 21, 3 (2004), 128 – 138. http://search.ebscohost.com.proxy.library.cmu.edu/login.aspx?direct=true&db=buh&AN=15059214&site=ehost-live

[14] Sergi Fàbregues and Marie-Hélène Paré. 2007. Charmaz, Kathy C. (2006). Constructing Grounded Theory: A Practical Guide Through Qualitative Analysis. *Papers* 86 (2007), 284,287. https://doaj.org/article/eef57104c2d14e7485b3afe5868bd7fe

[15] Gary Alan Fine. 2002. *Shared fantasy: Role playing games as social worlds*. University of Chicago Press.

[16] William Gaver, Andy Boucher, Sarah Pennington, and Brendan Walker. 2004. Cultural probes and the value of uncertainty. *interactions-Funology* 11, 5 (2004), 53–56.

[17] James Paul Gee. 2003. What video games have to teach us about learning and literacy. *Computers in Entertainment (CIE)* 1, 1 (2003), 20–20.

[18] Erving Goffman. 2002. Front and Back Regions of Everyday Life [1959]. *The Everyday Life Reader. London, Routledge* (2002), 50–57.

[19] Steve R Harrison and Paul Dourish. 1996. Re-place-ing space: The roles of place and space in collaborative systems.. In *CSCW*, Vol. 96. 67–76.

[20] Li He. 2013. Couple collaboration: a design research exploration. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems (CHI EA '13)*, Vol. 2013-. ACM, 2689,2694.

[21] Julie A Kientz, Shwetak N Patel, Brian Jones, ED Price, Elizabeth D Mynatt, and Gregory D Abowd. 2008. The georgia tech aware home. In *CHI'08 extended abstracts on Human factors in computing systems*. ACM, 3675–3680.

[22] Maria Lewicka. 2011. Place attachment: How far have we come in the last 40 years? *Journal of environmental psychology* 31, 3 (2011), 207–230.

[23] Shelley Mallett. 2004. Understanding home: a critial review of the literature. *The Sociological Review* 52, 1 (2004), 62,89. http://search.proquest.com/docview/199283786/

[24] D. Massey and P. Jess. 1995. *A Place in the World?: Places, Cultures and Globalization*. Oxford: Oxford University Press. 45–85 pages.

[25] Tuuli Mattelmäki and others. 2006. *Design probes*. Aalto University.

[26] David HJ Morgan and DHJ Morgan. 1996. *Family connections: An introduction to family studies*. Vol. 45. Polity Press Cambridge.

[27] Aske Mottelson and Kasper Hornbæk. 2017. Virtual reality studies outside the laboratory. In *Proceedings of the 23rd acm symposium on virtual reality software and technology*. ACM, 9.

[28] David M Newman and Elizabeth Grauerholz. 2002. *Sociology of families*. Pine Forge Press.

[29] William Odom, John Zimmerman, Jodi Forlizzi, Hajin Choi, Stephanie Meier, and Angela Park. 2012. Investigating the presence, form and behavior of virtual possessions in the context of a teen bedroom. In *Proceedings of the SIGCHI Conference on human factors in computing systems (CHI '12)*. ACM, 327,336.

[30] Virpi Oksman, Sanni Siltanen, and Mari Ainasoja. 2012. User participation in co-creative services: developing virtual and augmented reality tools for do-it-yourself home design. In *Proceeding of the 16th International Academic MindTrek Conference (MindTrek '12)*. ACM, 229,230.

[31] Doenja Oogjes, William Odom, and Pete Fung. 2018. Designing for an Other Home: Expanding and Speculating on Different Forms of Domestic Life. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18)*. ACM, New York, NY, USA, 313–326. DOI: http://dx.doi.org/10.1145/3196709.3196810

[32] Chris Pereira. 2014. Sony's Virtual World, PlayStation Home, Closing Its Doors After 6 Years. (2014). https://www.gamespot.com/articles/sonys-virtual-world-playstation-home-closing-its-d/1100-6422590/

[33] Kathryn E Ringland. 2019. A place to play: the (dis) abled embodied experience for autistic children in online spaces. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.

[34] Eugene Rochberg-Halton. 1981. *The meaning of things: Domestic symbols and the self*. Cambridge [Eng.]; New York: Cambridge University Press.

[35] Johnny. Saldaña. 2013. *The coding manual for qualitative researchers* (2nd ed. ed.). SAGE, Los Angeles.

[36] Margrit Schreier. 2012. *Qualitative content analysis in practice*. Sage Publications.

[37] Abigail Sellen, Richard Harper, Rachel Eardley, Shahram Izadi, Tim Regan, Alex S Taylor, and Ken R Wood. 2006. HomeNote: supporting situated messaging in the home. In *Proceedings of the 2006 20th anniversary conference on Computer supported cooperative work*. ACM, 383–392.

[38] Sally A Shumaker and Ralph B Taylor. 1983. Toward a clarification of people-place relationships: A model of attachment to place. *Environmental psychology: Directions and perspectives* 2 (1983), 19–25.

[39] Adalberto L Simeone and Eduardo Velloso. 2015. Substitutional reality: bringing virtual reality home. *XRDS: Crossroads, The ACM Magazine for Students* 22, 1 (2015), 24–29.

[40] Peter Somerville. 1997. THE SOCIAL CONSTRUCTION OF HOME. *Journal of Architectural and Planning Research* 14, 3 (1997), 226–245. http://www.jstor.org/stable/43030210

[41] Misha Sra, Aske Mottelson, and Pattie Maes. 2018. Your Place and Mine: Designing a Shared VR Experience for Remotely Located Users. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18)*. ACM, New York, NY, USA, 85–97. DOI: http://dx.doi.org/10.1145/3196709.3196788

[42] Laurel Swan and Alex Taylor. 2008. Photo displays in the home. In *Proceedings of the 7th ACM conference on designing interactive systems (DIS '08)*. ACM, 261,270.

[43] A.S. Taylor and L. Swan. 2005. Artful systems in the home. *CHI 2005: Technology, Safety, Community: Conference Proceedings - Conference on Human Factors in Computing Systems* (2005), 641,650.

[44] Alex S Taylor, Richard Harper, Laurel Swan, Shahram Izadi, Abigail Sellen, and Mark Perry. 2007. Homes that make us smart. *Personal and Ubiquitous Computing* 11, 5 (2007), 383–393.

[45] Zachary O Toups, Nicole K Crenshaw, Rina R Wehbe, Gustavo F Tondello, and Lennart E Nacke. 2016. " The Collecting Itself Feels Good" Towards Collection Interfaces for Digital Game Objects. In *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play*. 276–290.

[46] Jonathan Vanian. 2016. Netflix Brings Its New Virtual Reality App to Google Daydream. (2016). http://fortune.com/2016/12/12/netflix-virtual-reality-google/

[47] Jayne Wallace, John McCarthy, Peter C Wright, and Patrick Olivier. 2013. Making design probes work. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 3441–3450.

[48] Salu Ylirisku, Antti Jylhä, Anu Lehtiö, Imtiaj Ahmed, Craig Stewart, Abigail Sellen, Richard Harper, and Giulio Jacucci. 2016. Designing for active place presence at home: The hole in space design experiment. In *Proceedings of the 2016 ACM Conference on Designing Interactive Systems*. ACM, 1341–1352.

[49] Hao Zhang, Qiong Wu, Chunyan Miao, Zhiqi Shen, and Cyril Leung. 2019. Towards Age-friendly Exergame Design: The Role of Familiarity. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play*. 45–57.